

Answers

1. True/False: Lerp stands for linear extrapolation.

Answer: False, linear "interpolation"

2. True/False: In Unity, the function for linear interpolation is referred to as "mix".

Answer: False, GLSL uses the name "mix"

3. True/False: Linear interpolation involves estimating unknown values between 2 known values

Answer: True

4. True/False: The formula for linear interpolation is $b * t + a * (1 - t)$.

Answer: True, although there's a shorter version $a + t * (b - a)$

5. True/False: The parameter 't' in lerp should be between 0 and 1.

Answer: True

6. True/False: Lerp can only be used in 1-dimensional problems.

Answer: False

7. True/False: InverseLerp is a standard function found in all programming languages and libraries.

Answer: False, many game development libraries provide it, but it's not guaranteed.

8. True/False: The clamp function in the inverseLerp implementation ensures the value of 't' remains between 0 and 1.

Answer: True

9. True/False: For $t = 0.0$ in the 1D case with $A = 10$ and $B = 20$, the result of $\text{lerp}(A, B, t)$ is A

Answer: True

10. True/False: Remap is used to convert a value from its original range to a target range.

Answer: True

11. True/False: The remapping function can be derived using the concept of inverse lerp and lerp combined.

Answer: *True*

12. Calculate lerp for A = 5, B = 15, and t = 0.4.

Answer:
$$\begin{aligned} & \text{lerp}(A, B, 0.4) \\ &= 5 + 0.4 * (15 - 5) \\ &= 9 \end{aligned}$$

13. Calculate lerp for A = 0 and B = 100, and t = 0.8.

Answer:
$$\begin{aligned} & \text{lerp}(A, B, 0.8) \\ &= 0 + 0.8 * (100 - 0) \\ &= 80 \end{aligned}$$

14. Calculate inverseLerp for minValue = 0, maxValue = 30, and value = 20.

Answer:
$$\begin{aligned} & \text{inverseLerp}(0, 30, 20) \\ &= (20 - 0) / (30 - 0) \\ &= 0.66 \end{aligned}$$

15. Calculate inverseLerp for minValue = 10, B maxValue = 50, and value = 40.

Answer:
$$\begin{aligned} & \text{inverseLerp}(10, 50, 40) \\ &= (40 - 10) / (50 - 10) \\ &= 30 / 40 \\ &= 0.75 \end{aligned}$$

16. Remap a value of 5 from the range [0, 10] to [50, 100].

Answer:
$$\begin{aligned} & \text{remap}(0, 10, 50, 100, 5) \\ \\ & t = \text{inverseLerp}(0, 10, 5) \\ & t = 0.5 \\ \\ & \text{result} = \text{lerp}(50, 100, 0.5) \\ & \text{result} = 75 \end{aligned}$$

17. In a video game, a player's health ranges from 0 to 200. When designing the UI, this health is represented by a bar that can have a width ranging from 0 to 400 pixels. If the player has a health of 150, what should be the width of the health bar?

Answer: `remap(0, 200, 0, 400, 150)`

`t = inverseLerp(0, 200, 150)`

`t = 0.75`

`result = lerp(0, 400, 0.75)`

`result = 300`

18. Remap a temperature value of 25°C from the range [-10°C, 40°C] to a display range of [0, 255] for an 8-bit color scale representation.

Answer: `remap(-10, 40, 0, 255, 25)`

`t = inverseLerp(-10, 40, 25)`

`t = 0.7`

`result = lerp(0, 255, 0.7)`

`result = 179 (178.5)`